

Session 20 Quiz - AI Bias & Fairness

Track B | Grades 9–12 | 20 Questions

Q1 [Type: MCQ | Difficulty: Easy | QID : 54880]

AI _____ happens when an artificial intelligence system produces results that are systematically unfair to certain groups of people - not as a random glitch, but as a pattern built into how the system operates.

- A. Error
- B. Hallucination
- C. Compression
- D. Bias

Correct Answer: D

Explanation: The key word is "systematically" - Bias is not a one-time mistake, it is built into the way the AI operates and keeps happening over and over. Errors and hallucinations are different concepts, and compression is unrelated.

Q2 [Type: MCQ | Difficulty: Easy | QID : 54881]

_____ is a mitigation strategy that involves intentionally collecting training data that genuinely reflects the diversity of people who will use the AI - including groups that are usually underrepresented.

- A. Algorithmic compression
- B. Disaggregated metrics
- C. Diverse Data
- D. Voice cloning

Correct Answer: C

Explanation: Diverse Data is the first mitigation strategy applied - before training - because most bias problems are easier to prevent at the data stage than to fix after deployment.

Q3 [Type: MCQ | Difficulty: Medium | QID : 54884]

For high-stakes decisions like hiring, loans, and healthcare, the recommended mitigation strategy is _____ - requiring a human to review the AI's recommendation before any final decision is made.

- A. Human-in-the-Loop
- B. Algorithm Bias
- C. Disaggregated Metrics
- D. Voice cloning

Correct Answer: A

Explanation: Human-in-the-Loop is the non-negotiable safeguard for life-affecting decisions. Only humans can apply context, empathy, and ethical reasoning to decide what is truly fair - which is why high-stakes AI is always meant to assist, never to decide alone.

Q4 [Type: MCQ | Difficulty: Medium | QID : 54887]

Bias testing is a one-time step: once an AI passes its bias audit before launch, no further testing is needed.

- A. True
- B. False

Correct Answer: B (False)

Explanation: "Ongoing Testing" is a named mitigation strategy specifically because bias can re-emerge as data, users, and contexts change. Each new version of a model should be tested again, and audits should be conducted regularly - not just once before launch. A model can pass a launch audit and still drift into bias as it sees new data.

Q5 [Type: MCQ | Difficulty: Easy | QID : 54888]

AI bias is usually the result of developers intentionally trying to discriminate against certain groups.

- A. True
- B. False

Correct Answer: B (False)

Explanation: Bias is not intentional. AI has no feelings or intentions. Bias comes from patterns in data and from often-unconscious design choices made by humans. AI is a pattern-matching engine that reflects the world it was trained on - including that world's existing unfairness. The harm is real, but the intent usually is not malicious - which is what makes bias hard to fix.

Q6 [Type: MCQ | Difficulty: Medium | QID : 54890]

An AI system reporting 95% overall accuracy can still be performing badly for specific demographic groups, because the overall number can hide major disparities between groups.

- A. True
- B. False

Correct Answer: A (True)

Explanation: An overall accuracy number can hide major disparities between groups. The facial recognition case is the canonical example: less than 1% error for lighter-skinned men, but over 34% error for darker-skinned women - yet "overall" the system could still look competent. The right tool is disaggregated metrics: break performance down by group. "You cannot fix what you do not measure."

Q7 [Type: MCQ | Difficulty: Easy | QID : 54892]

Three main sources of AI bias have been identified. Which of the following is one of those three sources?

- A. Network latency bias
- B. Storage bias
- C. Data Bias
- D. Bandwidth bias

Correct Answer: C

Explanation: The three named sources are Data Bias, Algorithm Bias, and Human Bias. Data bias happens when training data over- or under-represents groups; algorithm bias happens when the learning method latches onto unfair patterns; human bias enters through design choices, labels, and what gets measured.

Q8 [Type: MCQ | Difficulty: Medium | QID : 54895]

A voice assistant has trouble understanding speakers with regional dialects because the training recordings came almost entirely from one metropolitan area. Which source of bias is this?

- A.** Data Bias
- B.** Algorithm Bias
- C.** Human Bias
- D.** Network Bias

Correct Answer: A

Explanation: The system never heard those dialects during training, so it cannot recognise them - a clear case of underrepresented data. This is the exact pattern from the Mini Activity exercises in the material.

Q9 [Type: MCQ | Difficulty: Easy | QID : 54906]

Which of the following is a Bias MITIGATION strategy?

- A.** Hiding the AI's limitations from users to maintain trust and prevent them from doubting the system's outputs
- B.** Releasing the AI to the public faster in order to gather more user feedback and identify bias issues after deployment
- C.** Removing all underrepresented groups from the training data so the model can focus on the majority population
- D.** Transparency - being open about what data was used, how the AI was trained, and where it might be biased

Correct Answer: D

Explanation: The four mitigation strategies are Diverse Data, Ongoing Testing, Human-in-the-Loop, and Transparency. Transparency builds trust and accountability because it lets users, auditors, and policymakers see what assumptions and limitations are baked into a system.

Q10 [Type: MCQ | Difficulty: Medium | QID : 54908]

Which bias DETECTION technique involves breaking down overall performance numbers by group - for example, reporting accuracy separately for Group A, Group B, and Group C, instead of just one combined score?

- A. Cross-Validation Testing
- B. Performance Testing
- C. Disaggregated Metrics
- D. Transparency

Correct Answer: C

Explanation: Disaggregated metrics are the antidote to misleadingly high "overall" accuracy numbers. By reporting accuracy by group, you can see disparities that a single combined number would hide - exactly the gap the Buolamwini study exposed in facial recognition.

Q11 [Type: MCQ | Difficulty: Medium | QID : 54911]

The healthcare algorithm case study showed bias because the algorithm was designed to predict future healthcare COSTS rather than future ILLNESS. Why did that design choice cause harm?

- A. Predicting healthcare costs is computationally much harder than predicting illness, so the model produced unreliable results across all patient groups
- B. Lower historical spending on Black patients (due to systemic barriers) was misread by the algorithm as better health, underestimating their needs
- C. The model used disaggregated metrics that broke down performance by patient demographic, which introduced noise into the predictions
- D. The model predicted the wrong year of healthcare data and applied outdated cost patterns to current patients

Correct Answer: B

Explanation: The variable the model was trained to predict (cost) did not mean what designers assumed it meant (need). A patient's healthcare costs reflect not just their illness but also barriers to access - so by predicting cost, the model inherited those barriers as part of its definition of "healthy."

Q12 [Type: MCQ | Difficulty: Hard | QID : 54917]

A school district is installing facial recognition for library entry. They want to ensure the system works equally well for all students regardless of skin tone, hair style, or facial features. Which mitigation strategy should they apply FIRST, before deployment?

- A.** Pause the entire facial recognition system permanently with no further development or planned deployment in the future
- B.** Transparency - write a public blog post about how the system works and what data it was trained on
- C.** Speed up deployment immediately so the system can start collecting real-world data from students to improve later
- D.** Diverse Data - collect images of students with varied appearances, lighting conditions, and angles

Correct Answer: D

Explanation: The problem is preventable at the data stage: gathering balanced, diverse training examples is the strategy that addresses the cause directly. Other strategies (testing, transparency, HITL) matter too - but the first move before launch is to ensure the model has seen the people it will be used on.

Q13 [Type: MCQ Scenario | Difficulty: Medium | QID : 54919]

Scenario: A 2018 study by researcher Joy Buolamwini tested commercial facial-analysis systems and found error rates of less than 1% for lighter-skinned men but over 34% for darker-skinned women. Investigators discovered that the benchmark training datasets were over 77% male and over 83% white.

Question: Which source of bias does this case most clearly illustrate?

- A.** Data Bias
- B.** Algorithm Bias
- C.** Human Bias only
- D.** Network Bias

Correct Answer: A

Explanation: When the training dataset is overwhelmingly one group, the model becomes excellent at that group and unreliable for everyone else. The numbers (77% male, 83% white) make this a textbook case of data bias.

Q14 [Type: MCQ Scenario | Difficulty: Medium | QID : 54921]

Scenario: A loan model uses an algorithm that finds the strongest statistical pattern in repayment data. The strongest pattern turns out to be zip code, because zip code correlates well with historical repayment. The algorithm latches onto zip code even though it was not told to consider geography - because mathematically, that's the most efficient predictor.

Question: Which source of bias does this case most clearly illustrate?

- A.** Sampling-rate bias
- B.** Data Bias
- C.** Algorithm Bias
- D.** Storage bias

Correct Answer: C

Explanation: This is Algorithm Bias, not Data Bias or Human Bias. The training data was balanced and the developers deliberately left zip code out - so neither the data nor a human design choice introduced the geographic signal. The harm came from the algorithm itself: it was designed to find the strongest statistical pattern, and it latched onto a combination of features that effectively recreated zip code, because that combination was mathematically efficient. The algorithm's own optimisation behaviour produced the unfair outcome. The fix is not more data and not a different feature set - it is changing how the algorithm learns, for example by adding fairness constraints that penalise the model when its predictions correlate with protected attributes.

Q15 [Type: MCQ Scenario | Difficulty: Medium | QID : 54923]

Scenario: Developers building an AI tutoring app decide to use word problems based on baseball - a sport popular in their team's home country - to teach math concepts. After launch, students in regions where baseball is unfamiliar perform significantly worse, not because they cannot do the math, but because they get stuck decoding the unfamiliar context.

Question: Which source of bias does this case most clearly illustrate?

- A.** Algorithm Bias
- B.** Human Bias
- C.** Algorithm Bias
- D.** Storage bias

Correct Answer: B

Explanation: The choice of what to predict (cost) was the silent design assumption that broke fairness. Cost reflects history, not health - and that history included unequal access. Even with no bad intent, the proxy choice produced large-scale harm.

Q16 [Type: MCQ Scenario | Difficulty: Hard | QID : 54926]

Scenario: A wildlife monitoring AI was trained only on photos taken in North American national parks. A research team now wants to use it in a rainforest reserve on another continent - where the lighting, foliage, and animal species are very different.

Question: Which mitigation strategy should the team apply FIRST?

- A.** Performance Testing - run the AI on a sample of local photos to measure accuracy before deciding whether to retrain
- B.** Deploy the AI immediately and trust the predictions, since the model was already trained on a large dataset of wildlife photos
- C.** Skip testing because the AI was already validated once on North American photos and there is no need to repeat the process
- D.** Publish a transparency report about the AI's training data instead of testing it on the new rainforest environment

Correct Answer: A

Explanation: When a model trained for one environment is moved to a very different one, you cannot assume past accuracy still holds. Performance testing across new groups (here, new species and conditions) is the bias-detection technique that surfaces problems before they cause harm. Diverse data may eventually be needed, but testing is the essential first step.

Q17 [Type: MCQ Scenario | Difficulty: Medium | QID : 54928]

Scenario: A housing agency is reviewing a new AI tool that screens rental applications. Before deployment, internal testing shows that the AI rejects applicants from certain postal codes at a 20% higher rate than other applicants.

Question: Which mitigation strategy should the housing agency apply BEFORE deciding whether to deploy?

- A.** Diverse Data only
- B.** Faster deployment to gather more real-world data
- C.** Disable all tools and never use AI again
- D.** Human-in-the-Loop

Correct Answer: D

Explanation: A 20% disparity on a high-stakes decision (housing) is a serious red flag. Humans must investigate the cause before this tool affects real applicants. Deploying first and asking questions later is exactly what causes documented harm.

Q18 [Type: MCQ Scenario | Difficulty: Hard | QID : 54930]

Scenario: Developers building a loan-approval model decide to include "proximity to a major transit station" as a factor - they reason that close access to transit might relate to job opportunities. After deployment, they discover the model is unintentionally disadvantaging financially comparable applicants who happen to live in rural areas.

Question: Which source of bias does this case most clearly illustrate?

- A. Data Bias
- B. Algorithm Bias
- C. Human Bias
- D. Network Bias

Correct Answer: C

Explanation: The training data itself may have been fine. The Bias entered through a human decision - choosing a feature that turned out to correlate unfairly with geography rather than with what the model was supposed to measure. Even well-intentioned feature choices can introduce harm.

Q19 [Type: FRQ - Short Answer | Difficulty: Hard | QID : 54933]

Scenario: A social media company's automatic photo-tagging feature mislabels - or fails to label - photos of people with darker skin tones. Searches for "outdoor activity" return mostly images of lighter-skinned people, while similar photos of darker-skinned people are tagged with generic or less descriptive labels. The company's engineers investigate and find that the training dataset contained far fewer images of people with diverse skin tones.

Question: Identify the PRIMARY source of bias in this case AND describe two concrete mitigation strategies the company could apply before relaunching the feature.

(Respond in 3–4 sentences.)

Model Answer: The primary source is Data Bias - the training dataset lacked sufficient images of people with diverse skin tones, so the model never learned to label them accurately. The first mitigation strategy is Diverse Data: intentionally collect a balanced dataset with people of varied skin tones in many activities and lighting conditions, ensuring equal representation. The second strategy is Ongoing Testing using disaggregated metrics: after retraining, run the tagger on photos grouped by skin tone and report accuracy separately for each group rather than relying on a single overall accuracy score. Together these address the cause directly and create a way to catch problems if they re-emerge in future versions.

Q20 [Type: FRQ - Scenario Analysis | Difficulty: Hard | QID : 54935]

Scenario: A hospital is preparing to deploy an AI system that helps doctors decide which patients need follow-up scans after an emergency room visit. The hospital wants to make sure the system is fair from day one.

Question: Describe how the hospital could apply ALL FOUR mitigation strategies from this material - Diverse Data, Ongoing Testing, Human-in-the-Loop, and Transparency - to this specific AI system. Be concrete about what each strategy would look like in practice.

(Respond in 3–4 sentences.)

Model Answer: For Diverse Data, the hospital should ensure its training set includes patients across age, gender, race, and socio-economic backgrounds, and avoid using a proxy like past spending that could hide unequal access - the lesson from the healthcare algorithm case. For Ongoing Testing, it should regularly run disaggregated performance tests, measuring follow-up recommendation accuracy separately for each demographic group, and re-test whenever the model is updated. For Human-in-the-Loop, every AI recommendation should be reviewed by a doctor before any final decision, since follow-up scans are a life-affecting choice and AI alone should never make that call. For Transparency, the hospital should publish documentation explaining what data the model was trained on, how it was tested, and where its known limitations are - so patients, doctors, and outside auditors can hold the system accountable. Together these four strategies turn fairness from a one-time check into an ongoing practice.